Technical Document #191: Retrieval Augmented Generation - Comprehensive Overvi

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Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Key Takeaways:

- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.
- Hybrid search combines keyword-based and semantic search.
- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.